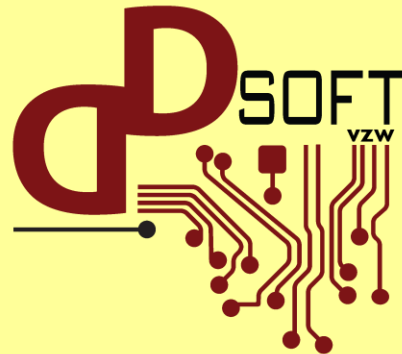


Making enabling apps for disabled people

APPS WITH DATA FOR PEOPLE WITH A DISABILITY

WITH XAMARIN FORMS



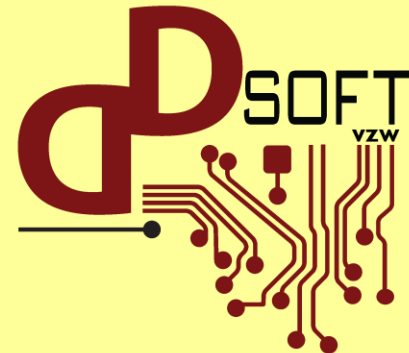
Who am I



Dennie Declercq.
Founder of DDSOFT vzw.
Volunteer developer in vzw Ithaka.

Obsessed about using technologies to make people with a disability their life better.
Maybe because I have my own disability (Autism).

Honored to speak here!



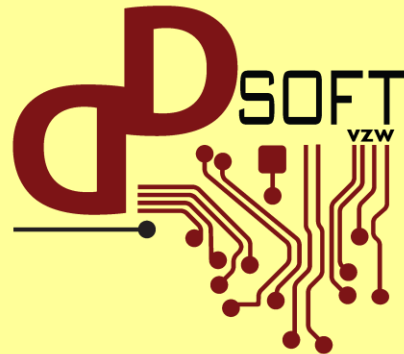
Technology we use

To show all the things, we create a simple Xamarin Forms project.

For practical reasons, we build only for Android at this time!



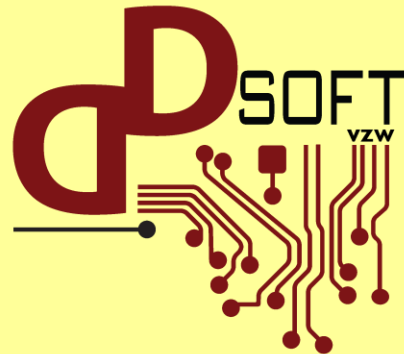
Xamarin



What this session is not about

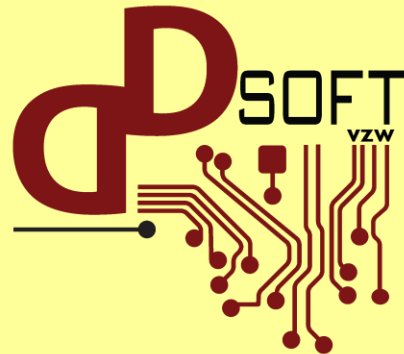
This session is not a deep dive into building apps in Xamarin.

I assume at least that you are common with creating basic apps in Xamarin.



What this session is about

This session is to let you know, how you can write apps for a new user-base with programming techniques we all know.



Can you realize ?

Vrijheid is voor Geert Wilders een privilege

Ondankbare taak van rechter om alles te wegen

Wilders wil een Koranverbod, maar bepleit voor zichzelf onbeperkte vrijheid van meningsuiting. Hypocriet, stelt **Marloes van Noorloos**.

Velen zullen een zucht van verlichting hebben geslaakt toe rechtbankvoorzitter Marcel van Oosten vanmorgen, zoals in een spannende thriller, Wilders' uitingen stuk voor stuk voorzag van het stempel 'vrijpraak'. De beschuldigingen aan het adres van de rechterlijke macht waren immers niet te overzien geweest als het college was gekomen tot een veroordeling. In Wilders' eigen opvatting zouden dan miljoenen mensen „rechtig geen vrouwen meer hebben in de rechterlijke macht“.

Er is geen twijfel over mogelijk dat hij alle mogelijkheden zou hebben aangegrepen om dat vertrouwen nog verder naar beneden te halen. Los van de voors en tegens van de uitspraak zelf is het zorgelijk dat de houdbaarheid van de rechtsstaat in Wilders' ogen staat of valt met de uitkomst van zijn eigen proces.

Toch is Wilders nog nooit principieel in het geweer gekomen tegen de wetgeving over haatuitingen. De politicus schreeuwt dan wel moord en brand over zijn eigen vervolging. In 2005 ondertekende zijn partij nog een motie die de autoriteiten oproep om meer werk te maken van de strafrechtelijke aanpak van haatuitingen. Tegenover een onbeperkte uitingenvrijheid voor zichzelf pleit Wilders juist voor het beperken van de vrijheid van meningsuiting van zijn tegenstanders, zoals het verbieden van de Koran en het aanpakken van radicaal-islamitische uitingen. De ui-

tingsvrijdeld werdvoet op die manier tot een 'privilege' in plaats van een recht dat juist geldt voor opvoeding en men verafschuwt. In zo'n klimaat komt de rechtsstaat – en de hoeders daarvan: de rechterlijke macht – onvermijdelijk onder vuur te liggen. Het is daarom van belang dat de ondankbare taak om die vermeende 'privileges' af te wegen tegen andere rechten en belangen. In dit geval is die belangenafweging in Wilders' voordeel uitgevallen, maar de rechter heeft op die gedachte laag zien dat het in feite alle kanten op had kunnen gaan.

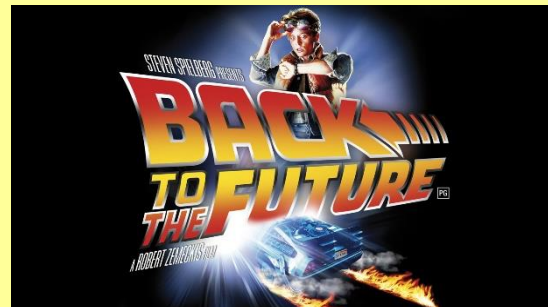
Laten we, nu de zaak is afgedaan, dan eens beginnen om de wettelijke

We moeten waken voor een te grote rol van het strafrecht

grenzen van de vrijheid van meningsuiting in Nederland eens goed tegen het licht te houden. Om welke redenen willen we bepaalde uitingen al dan niet verbieden? Onze wetgeving stamt deels uit de jaren dertig en deels uit de jaren zeventig. Zijn de ideeën die destijds relevant waren dat nu nog steeds? We moeten waken voor een al te grote rol voor het strafrecht. Een debat over de strafwaardigheid van Wilders' uitingen leidt immers af van waar het echt om gaat – of zijn ideeën ergens op slaan.

Marloes van Noorloos is promovenda aan het Willem Pompe Instituut voor Strafrechtswetenschappen van de Universiteit Utrecht. In december 2011 verschijnt haar proefschrift over haatuitingen.

*That the things that we're all do...
.. Are not common "life" for many people?*

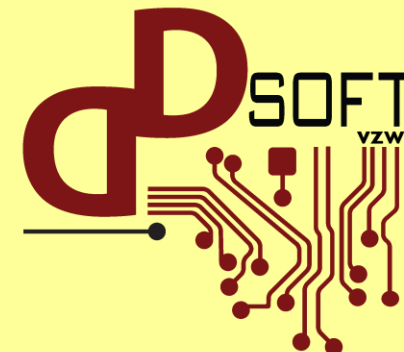


Reading a paper

Reading a subtitled movie

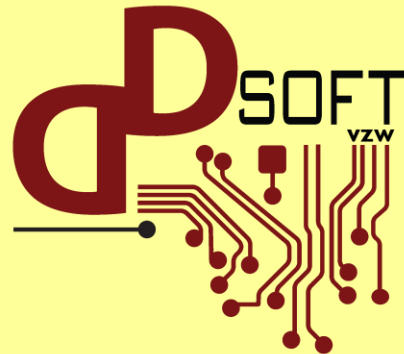


Ordering a pizza



There are **CA 5%** People with disability in Belgium

What means **CA 560.000** people cannot do everything like “we”



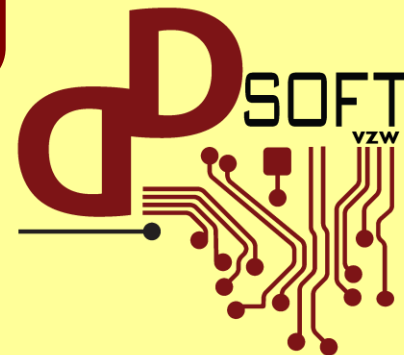
Two kinds of disabilities

Physical

- Can't move like the 'normal way'

Mental / Intellectual

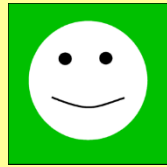
- Can't think / speak / read / behave (communicate) like the 'normal way'



What poorly exist

There are (very less) tech organisations who focus on people with physical disability's

And that's good!

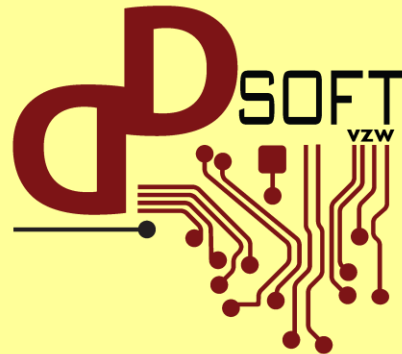


But there are quite less tech organisations who focus on people with mental / cognitive disabilities.

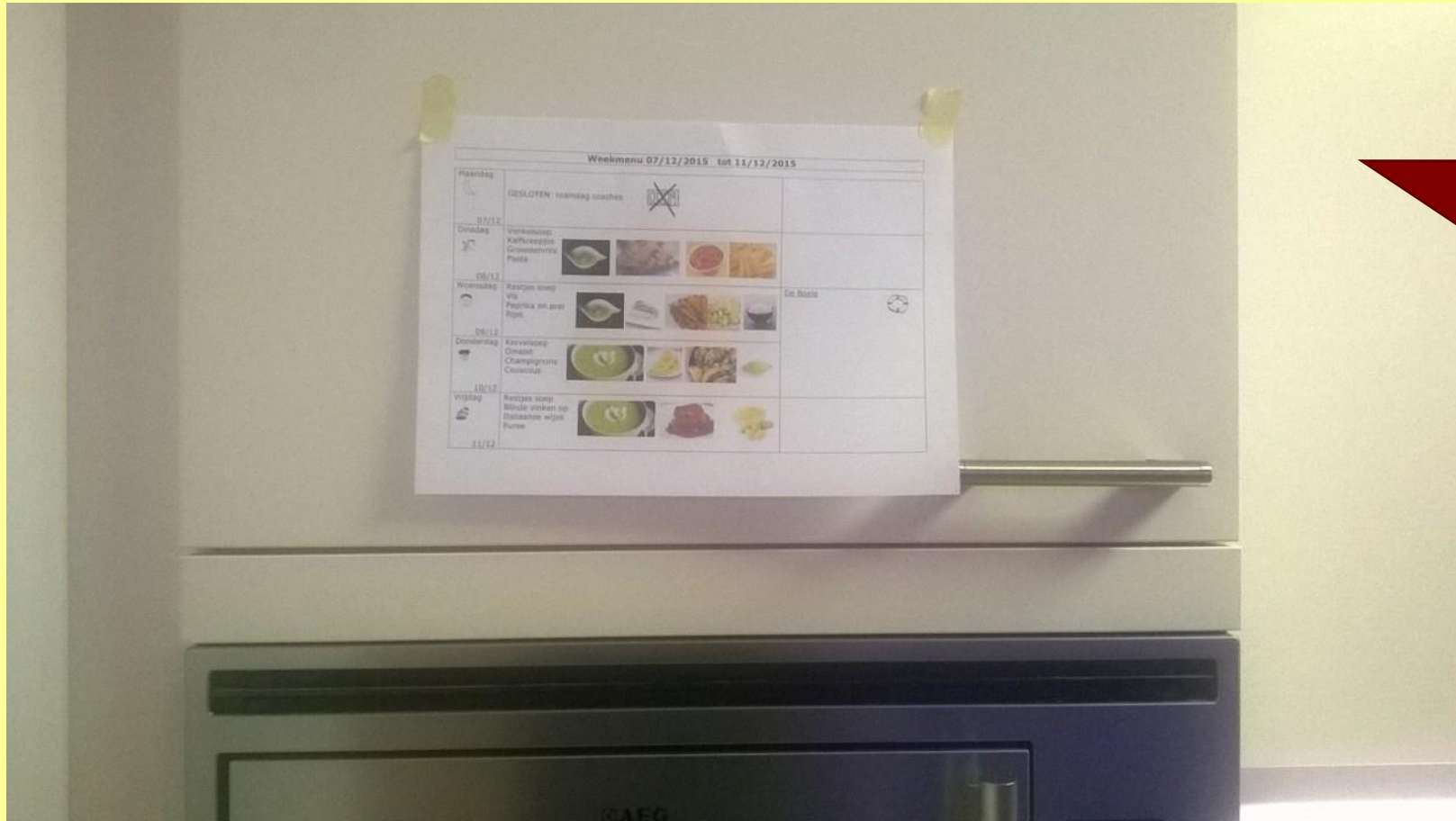
That's our (DDSoft nonprofit) goal.

But I hope...

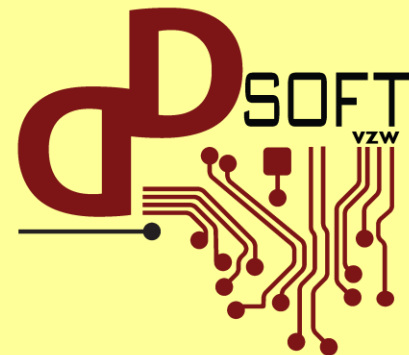
To inspire much more!



What's this?



An eating
schedule

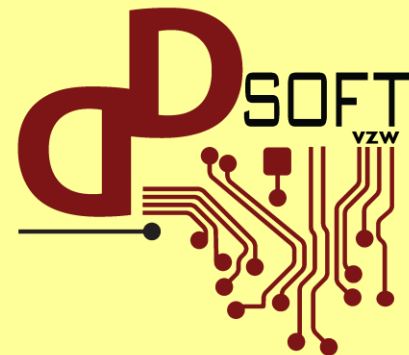


What's the purpose?

To let somebody, know what's to eat this week.

But what does that paper means,

if you can't read?

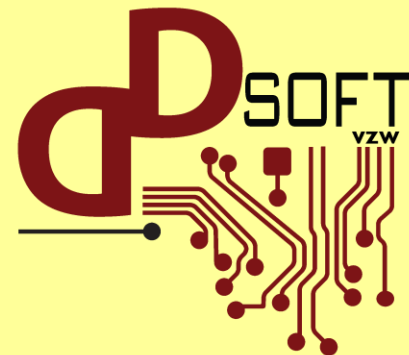
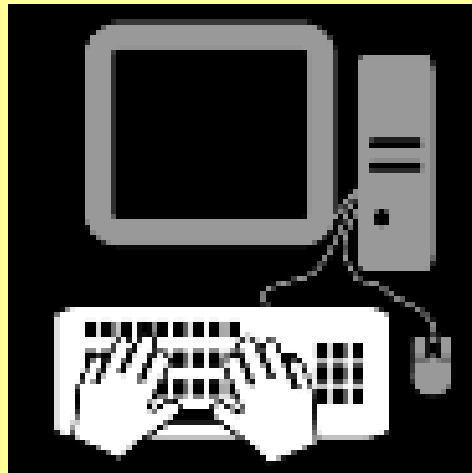


Where are developers for?

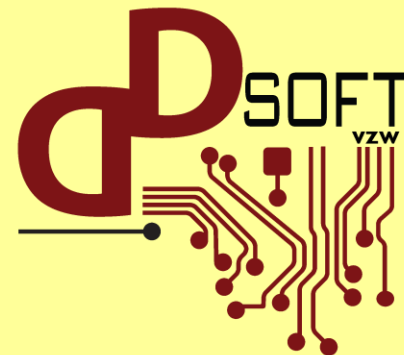
For making others' life easier.

To automate things that human cannot do,
or cannot do fast enough.

I hope you are all here for this reason!

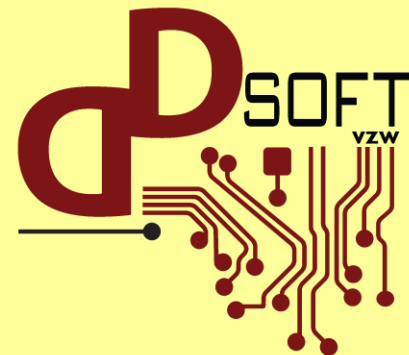


Let's learn to make apps for people with a disability



Three basic rules

1. Simplify the layout
2. Use pictographs or pictures
3. Use Text-To-Speech

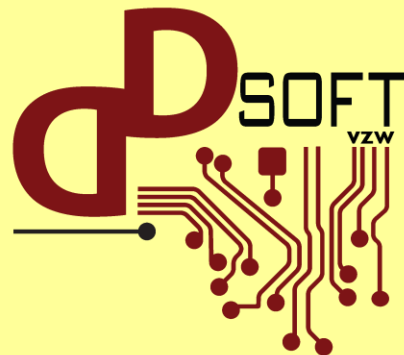
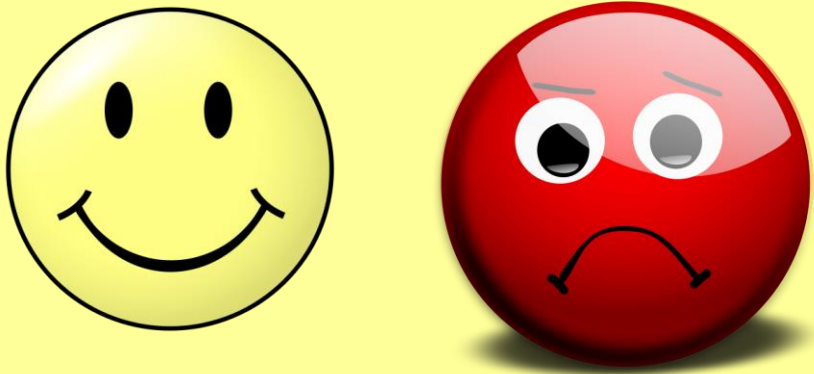


Demo Time 'Emoboard'!

Use case:

Let's assume you want to build an app to help them communicate about their mood.

Many people with a disability have difficulties to talk about their mood.

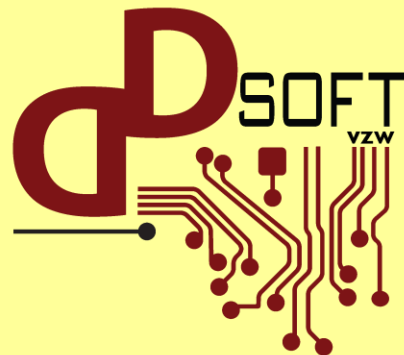


Demo Time – Starting point.

We start this demo with a solution, with the Xamarin Forms Template.

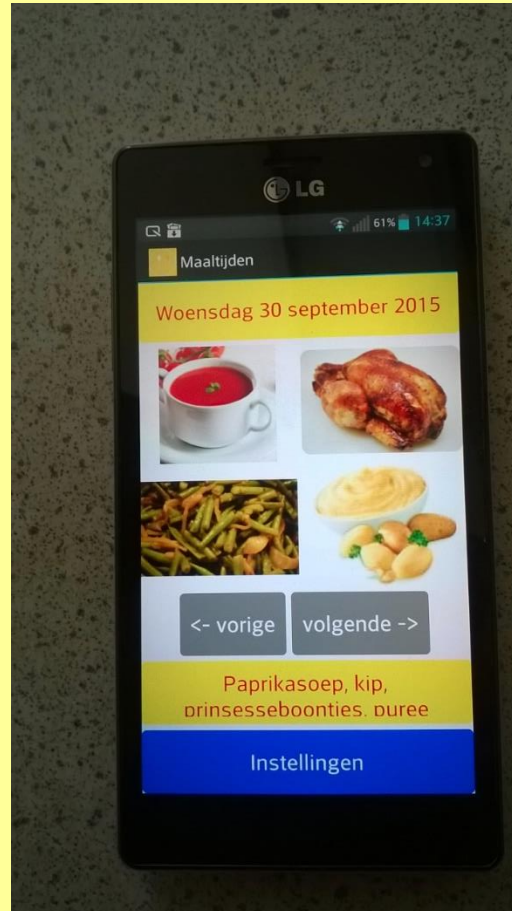
We have the file EmoBoard.cs

We create a MainPage



Simplify the layout

1. Use a simple white background color (nothings easier to read then white)
2. Use a few accent colors
3. Color by function
4. Be consistent!



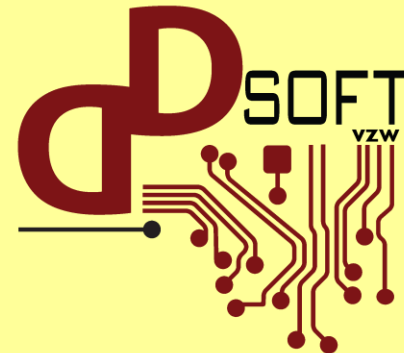
Color by function

I've created three button styles

I can speak !

I can navigate !

I open an other page!



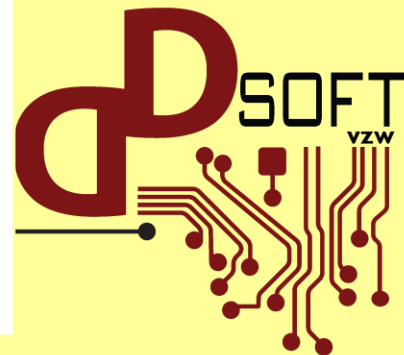
Color by funtion (in code)

```
public Button DayButton; // I CAN SPEAK  
  
public Button MealButton; // I CAN SPEAK  
  
public Button PrevButton; // I CAN NAVIGATE  
  
public Button NextButton; // I CAN NAVIGATE  
  
public Button SettingsButton; // I OPEN AN OTHER PAGE
```

```
DayButton = new Button {  
    BackgroundColor = Color.Yellow,  
    TextColor = Color.Red  
};
```

```
SettingsButton = new Button {  
    BackgroundColor = Color.Blue,  
    TextColor = Color.Yellow  
};
```

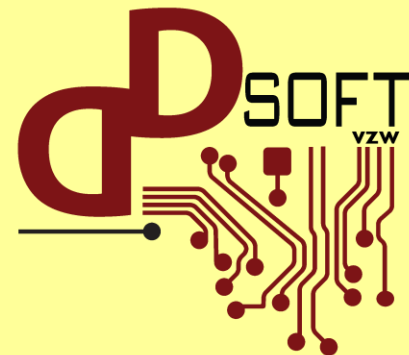
```
PrevButton = new Button{  
    BackgroundColor = Color.Gray,  
    TextColor = Color.White  
};
```



Demo

Demo Time

Making a low-barrier user interface for Happiness Indicator.



Use pictographs or pictures

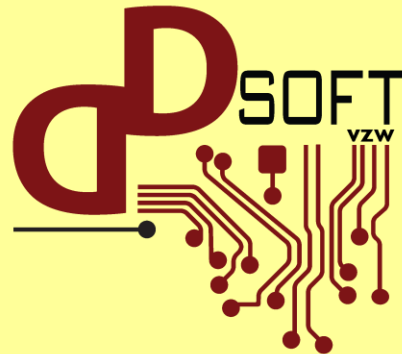
What means the following?

酢飯

The answer is:

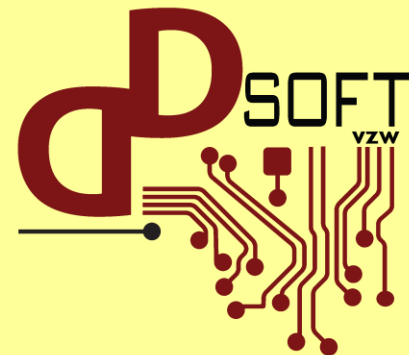
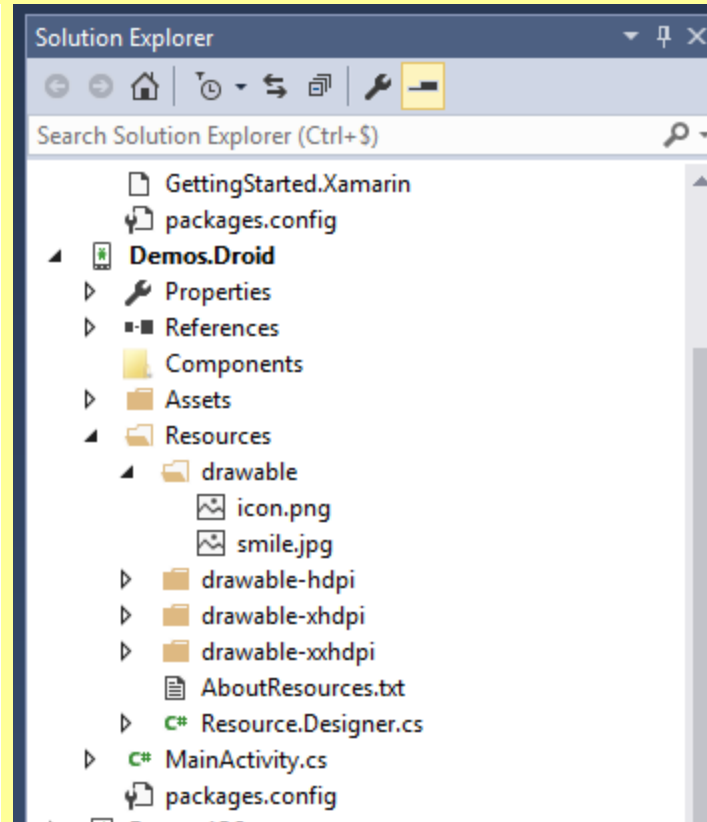


For some people with a disability,
who cannot read,
Letters and words means nothing.
Just like Chinese or Japanese
tokens for us.
Everybody (except people with
visual impairments) can see a
picture.



Use pictographs or pictures (in code)

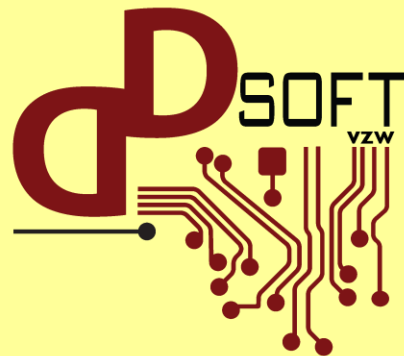
```
Image imgSmile = new Image  
{  
    Source = "smile.jpg"  
};
```



Demo

Demo Time!

Adding some pictographs and pictures.



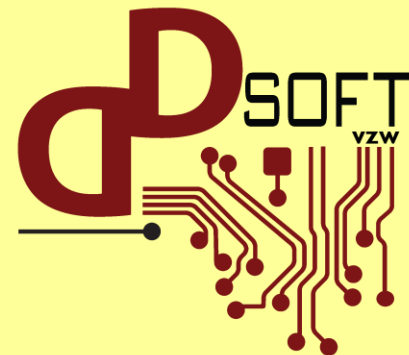
Use Text-To-Speech (TTS)

For some people, pictures aren't enough.

To reach them also, you can use TTS, Text-To-Speech

You have to do three steps:

- 1 Make a TextToSpeech Dependency Interface
- 2 Make implementation of that interface for each OS you want to implement (in this case only Android)
- 3 Call the implementation where needed



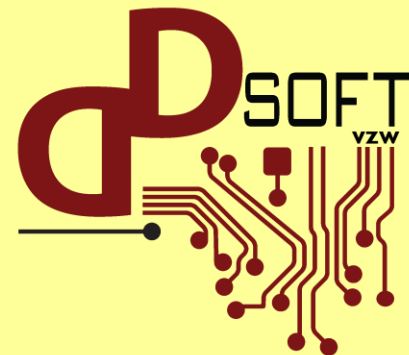
TTS – Making a dependency interface

In the shared PCL add a new c# interface

→ Add the speak function

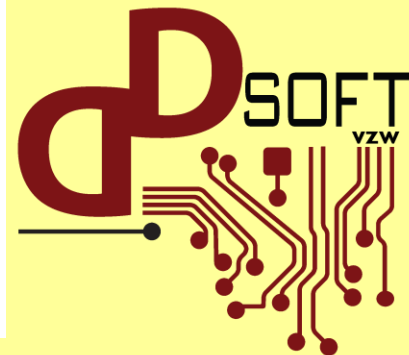
```
public interface ITextToSpeech
{
    void Speak(string text);
}
```

**** It's a best practice to make a folder specific for dependency interfaces ****



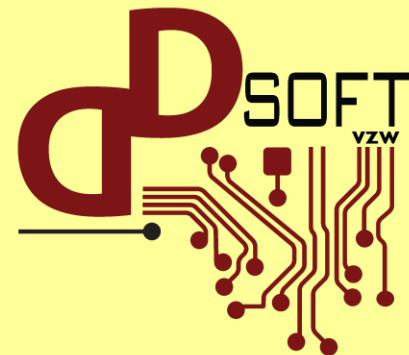
TTS – Making the Android implementation

```
public class TextToSpeechImplementation : Java.Lang.Object, ITextToSpeech,
TextToSpeech.IOnInitListener {
    TextToSpeech speaker;
    string toSpeak;
    public TextToSpeechImplementation() { }
    public void Speak(string tekst) {
        var ctx = Forms.Context;
        toSpeak = text;
        if (speaker == null) {
            speaker = new TextToSpeech(ctx, this);
        } else {
            var p = new Dictionary<string, string>();
            speaker.Speak(toSpeak, QueueMode.Flush, p);
        }
    }
}
```



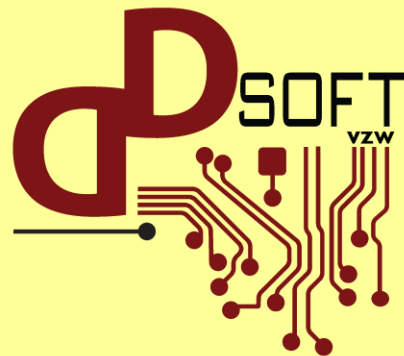
TTS – Call the speak function

```
void DayButton_Clicked(object sender, EventArgs e)
{
    string text = DayButton.Text.ToString();
    DependencyService.Get<ITextToSpeech>().Speak(text);
}
```



Demo

Demo of Text-To-Speech

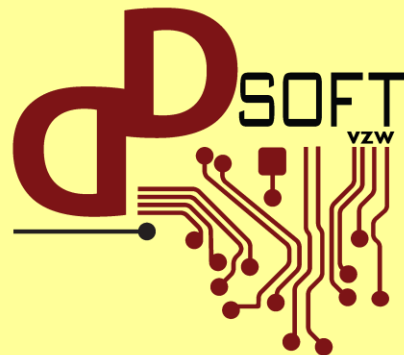


Recap

Today you have seen how you can make an app accessible for people who cannot read or write due cognitive disabilities.

You have learn to:

1. Simplify the layout with color by function
2. Use images and pictographs
3. Use TTS



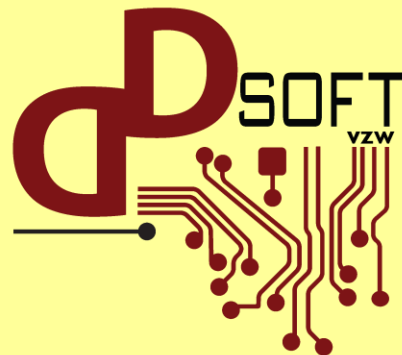
Now it's up to you

The journey of apps for people with a disability just starts here.

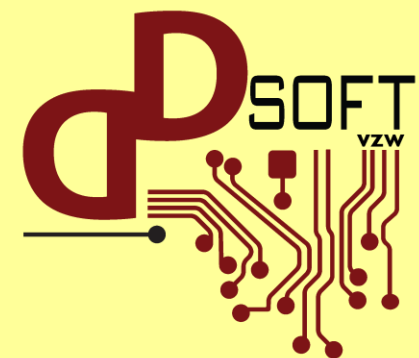
The next big thing is using data (like my own open data platform)

<http://socialdata.ddsoft.be/>

So you cannot only let them explain their feelings, but also let them inform about their life.



Q&A



Thanks!

